mergencystretch

Chapter 10. Causal Discovery

Constraint-based, score-based and functional methods; local causal discovery

Contents

10.1 Causal discovery **2**
10.1.1 The structure-learning problem . . . 2
10.1.1.1 DAG models . . . 2
10.1.1.2 Faithfulness . . . 2
10.1.2 Constraint-based structure learning . . . 3
10.1.2.1 What constraints a DAG encodes . . . 3
10.1.2.2 Markov equivalence . . . 3
10.1.2.3 The PC algorithm . . . 4
10.1.2.4 Extensions . . . 5
10.1.2.5 Summary . . . 5
10.1.3 Score-based approaches . . . 5
10.1.3.1 Why scores? . . . 5
10.1.3.2 Log-likelihood for DAG models . . . 6
10.1.3.3 BIC . . . 6
10.1.3.4 Search and GES . . . 6
10.1.3.5 Continuous optimization: NOTEARS and DAGMA . . . 7
10.1.3.6 Summaries . . . 8
10.1.4 Parametric approaches . . . 8
10.1.4.1 Non-linear functions . . . 9
10.1.4.2 Linear functions with non-Gaussian noise: LiNGAM . . . 9
10.1.4.3 Score matching (NoGAM) . . . 10
10.1.5 Appendix: GES operators in detail . . . 10
10.1.6 Appendix: Post-nonlinear causal model (optional) . . . 10
10.1.7 Appendix: Scores in detail . . . 11
10.1.8 Appendix: CI tests for causal discovery . . . 11

10.2 Local causal discovery **12**
10.2.1 The map . . . 12
10.2.1.1 One primitive: grow–shrink . . . 12
10.2.2 Going local on CI . . . 13
10.2.2.1 The Markov blanket . . . 13
10.2.2.2 GS: grow–shrink the blanket . . . 13
10.2.2.3 HITON: target-focused, not whole-graph . . . 14
10.2.3 Bringing in the score . . . 14
10.2.3.1 MMHC: a hybrid bridge . . . 14
10.2.3.2 BOSS: best order score search . . . 15
10.2.4 Zoom out . . . 15

10.1 Causal discovery

Motivation. The central question is how much of the causal diagram can be recovered from data alone, under which assumptions, and with which algorithms.

10.1.1 The structure-learning problem

Motivation. State what causal discovery takes as input and returns as output, and the assumptions needed.

Earlier lectures assumed the diagram $\mathcal{G}$ induced by an unknown SCM was given. Causal discovery goes the other way: from $P(\mathbf{V})$ (or samples) to $\mathcal{G}$. A cartoon: does the rooster's crow cause the sun to rise? A table of binary observations of (rooster, sun) shows them almost always equal, but the data alone do not say which way the arrow goes.

Structure learning. *Input:* a data set assumed sampled i.i.d. from $P(\mathbf{V})$. *Output:* a set of graphs consistent with what we know about the data (hopefully including $\mathcal{G}$). It is an unsupervised learning problem. Extensions: multiple data sets, observational and interventional data, unknown interventions, ...

10.1.1.1 DAG models

We focus on DAG models: a causal structure on $\mathbf{V}$ is a DAG with one node per variable, each directed edge representing a direct functional relationship. DAG models correspond to Markovian models, i.e. *causal sufficiency*: no unmeasured confounder of the measured variables. Three equivalent definitions, illustrated on $A \to B$, $A \to C$, $B \to D$, $C \to D$, $D \to E$:

- **Factorization:** $P(A,B,C,D,E) = P(E \mid D)P(D \mid B,C)P(C \mid A)P(B \mid A)P(A)$.
- **Local Markov property:** $(C \perp\!\!\!\perp B \mid A)$, $(D \perp\!\!\!\perp A \mid B,C)$, $(E \perp\!\!\!\perp A,B,C \mid D)$.
- **Global Markov property:** if $\mathbf{A}$ is d-separated from $\mathbf{B}$ by $\mathbf{C}$ in $\mathcal{G}$, then $\mathbf{A} \perp\!\!\!\perp \mathbf{B} \mid \mathbf{C}$ in $P(\mathbf{V})$.

10.1.1.2 Faithfulness

The global Markov property is one-directional: every d-separation gives a CI, equivalently every dependence gives a d-connection. A complete graph is compatible with every $P(\mathbf{V})$, so we need minimality, an assumption that reverses the implication.

Definition (Faithfulness)

For all $\mathbf{A},\mathbf{B},\mathbf{C}$: $\mathbf{A}$ is d-separated from $\mathbf{B}$ given $\mathbf{C}$ in $\mathcal{G}$ *iff* $\mathbf{A} \perp\!\!\!\perp \mathbf{B} \mid \mathbf{C}$ in $P(\mathbf{V})$.

Faithfulness is an extra property that a $P(\mathbf{V})$ factorizing according to $\mathcal{G}$ may satisfy. Most such distributions are faithful (unfaithful ones have measure zero). However: in finite samples unfaithful and nearly unfaithful distributions are hard to tell apart, and there are many more of the latter; nature does not pick distributions at random and may evolve unfaithful configurations (homeostasis, equilibria). In practice one should justify why faithfulness is sensible for the problem.

Example 10.1 (Violation by cancellation)

$X = \epsilon_X$, $Z = X + \epsilon_Z$, $Y = X - Z + \epsilon_Y$ (edges $X \xrightarrow{1} Z$, $Z \xrightarrow{-1} Y$, $X \xrightarrow{1} Y$). X and Y are not d-separated by $\emptyset$, yet $Y = X - (X + \epsilon_Z) + \epsilon_Y = -\epsilon_Z + \epsilon_Y$, so $X \perp\!\!\!\perp Y$: the two paths cancel

exactly.

10.1.2 Constraint-based structure learning

Motivation. Find the CIs that hold, rule out inconsistent graphs, and return what is left.

10.1.2.1 What constraints a DAG encodes

Example: $X \to W \to Y$, $X \to Z \leftarrow Y$. Of the candidate statements

$$(1)\ X \perp\!\!\!\perp Y \mid W, \quad (2)\ X \perp\!\!\!\perp Y, \quad (3)\ X \perp\!\!\!\perp Y \mid W, Z, \quad (4)\ X \perp\!\!\!\perp Y \mid Z,$$

only (1) is forced by the graph; (2)–(4) are dependences ($X \to W \to Y$ open; the collider Z opened). The data must exhibit the graph's independences for the graph to be valid.

With faithfulness, d-separation in $\mathcal{G} \Leftrightarrow$ CI in P, so CIs refine the possible DAGs. Without any CI information the best guess is the *complete undirected* graph (every pair adjacent, no orientation).

Adjacency. If for $A, B \in \mathbf{V}$ some $\mathbf{S} \subseteq \mathbf{V} \setminus \{A, B\}$ gives $A \perp\!\!\!\perp B \mid \mathbf{S}$, then A and B are not adjacent. (Under faithfulness, adjacent variables are dependent given every set.) In the example the only removed edge is X–Y (separated by $\{W\}$); the result, called the *skeleton*, is $X - W - Y$, $X - Z - Y$.

Orientation. For non-adjacent A, B there is a separator $\mathbf{S}$. If $A - C - B$ (an *unshielded triple*) and $C \notin \mathbf{S}$, orient $A \to C \leftarrow B$, an *unshielded collider* (v-structure); otherwise C is an unshielded non-collider (e.g. $A \to C \to B$). In the example the separator $\{W\}$ of X, Y does not include Z, so $X \to Z \leftarrow Y$; this rules out $X \to Z \to Y$, $X \leftarrow Z \leftarrow Y$ and $X \leftarrow Z \to Y$. Can anything else be oriented? $W \in \mathbf{S}$ makes $X - W - Y$ a non-collider, but its direction is undetermined.

10.1.2.2 Markov equivalence

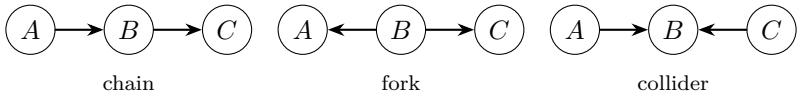


Figure. *The chain and the fork imply the same independence, $A \perp\!\!\!\perp C \mid B$, and are Markov equivalent; the collider implies $A \perp\!\!\!\perp C$ but $A \not\perp\!\!\!\perp C \mid B$. This is why observational data can orient unshielded colliders but not the other edges.*

The three DAGs $X \to W \to Y$, $X \leftarrow W \leftarrow Y$, $X \leftarrow W \to Y$ (each with $X \to Z \leftarrow Y$) entail the same independences: they are *observationally* (Markov) *equivalent*; the local Markov property gives the same CIs. A *Markov equivalence class* (MEC) is a set of Markov-equivalent DAGs.

Theorem 10.1 (Verma and Pearl 1990)

Two DAGs represent the same statistical model iff they have the same skeleton and the same unshielded colliders.

Proof [Supplement]. *If.* Two DAGs with the same skeleton and unshielded colliders have the same d-separations: a path is open given $\mathbf{S}$ depending only on which inner nodes are colliders and their relation to $\mathbf{S}$; for unshielded triples the collider status agrees by assumption, and for a shielded triple one can replace the path segment by the shielding edge, so a shortest open path has no shielded colliders whose status differs. Same d-separations means the same Markov (statistical) model. *Only if.* Different skeletons: an edge $A - B$ in one DAG and not the other; in the second A and B are separated by some set (e.g. the parents of the later one), but never in the first. Same

skeleton but an unshielded collider $A \to C \leftarrow B$ in one and not the other: in the first, A and B are separated by a set excluding C; in the second, every separating set must contain C (the path $A - C - B$ is open otherwise). In both cases the d-separations differ, and so do the models. ■

All DAGs in a class are represented by one mixed graph (directed and undirected edges), the *pattern* or CPDAG (completed partially directed acyclic graph). For three variables (Eberhardt 2017), the 25 DAGs fall into 11 classes; e.g. the chain, reversed chain and fork on X, Y, Z form one class, while the collider $X \to Y \leftarrow Z$ is alone in its class.

10.1.2.3 The PC algorithm

Basic idea:

(1) non-adjacent vertices have a separator;

(2) for an unshielded collider $X \to Z \leftarrow Y$, the separator must not contain Z.

Algorithm 1 PC algorithm (Spirtes and Glymour)

start with the complete undirected graph $\mathcal{G}$ on n vertices
for $i = 0, \dots, n-2$ **do** ▷ skeleton
  for every adjacent pair A, B in $\mathcal{G}$ **do**
    if $\exists \mathbf{S}$ with $|\mathbf{S}| = i$, $\mathbf{S} \subseteq \mathrm{adj}_{\mathcal{G}}(A) \cup \mathrm{adj}_{\mathcal{G}}(B) \setminus \{A, B\}$, and $A \perp\!\!\!\perp B \mid \mathbf{S}$ **then**
      remove $A - B$; sepset$[\{A, B\}] \leftarrow \mathbf{S}$
for every non-adjacent A, B with $A - C - B$ **do** ▷ v-structures
  if $C \notin$ sepset$[\{A, B\}]$ **then** orient $A \to C \leftarrow B$
orient as many remaining edges as possible without creating new v-structures or directed cycles (Meek's rules)

Meek's rules.

1. Orient $b - c$ as $b \to c$ if there is $a \to b$ with a, c non-adjacent (else a new v-structure $a \to b \leftarrow c$ would appear).
2. Orient $a - b$ as $a \to b$ if there is a chain $a \to c \to b$ (else a cycle).
3. Orient $a - b$ as $a \to b$ if there are two chains $a - c \to b$ and $a - d \to b$ with c, d non-adjacent (if $b \to a$, Rule 2 would force $c \to a$ and $d \to a$, a new v-structure at a).
4. Orient $a - b$ as $a \to b$ if there are two chains $a - c \to d$ and $c \to d \to b$ with c, b non-adjacent and a, d adjacent.

Example 10.2 (Walk-through)

True (unknown) DAG: $A \to B$, $A \to C$, $B \to D$, $C \to D$. CIs revealed by the tests: $B \perp\!\!\!\perp C \mid A$ (fork blocked by A, collider D closed) and $A \perp\!\!\!\perp D \mid \{B, C\}$ (both paths blocked).

- $|\mathbf{S}| = 0$: all pairs dependent; nothing removed.
- $|\mathbf{S}| = 1$: $B \perp\!\!\!\perp C \mid A$: remove $B - C$, sepset$[\{B, C\}] = \{A\}$; other tests dependent.
- $|\mathbf{S}| = 2$: $A \perp\!\!\!\perp D \mid \{B, C\}$: remove $A - D$, sepset$[\{A, D\}] = \{B, C\}$. Skeleton: $A - B$, $A - C$, $B - D$, $C - D$.
- V-structures: $B - D - C$ with $D \notin \{A\}$, so $B \to D \leftarrow C$; $B - A - C$ with $A \in \{A\}$, not a collider; $A - B - D$, $A - C - D$ with $B, C \in \{B, C\}$, not colliders.
- Meek's rules: Rule 1 applied to $A - B$ (using $B \to D$ with A, D non-adjacent) and to $A - C$, obtaining $A \to B$, $A \to C$, $B \to D \leftarrow C$, which matches the true DAG.

A caution on the last step: Rule 1 requires an arrow *into* B from a vertex non-adjacent to A, whereas $B \to D$ points out of B. Strictly, no Meek rule fires here, and the CPDAG is $A - B$, $A - C$, $B \to D \leftarrow C$. Its class contains $A \to B, A \to C$; $B \to A \to C$; $C \to A \to B$ (each with $B \to D \leftarrow C$), while $B \to A \leftarrow C$ is excluded because it would add a v-structure. The true DAG is one member of this class.

Exercise. Apply PC to $A \to B$, $A \to C$, $B \to D$, $C \to D$, $D \to E$ (the earlier five-variable DAG). The v-structure $B \to D \leftarrow C$ is found, and Rule 1 orients $D \to E$ (since $B \to D$ with B, E non-adjacent).

10.1.2.4 Extensions

Conservative PC (Ramsey, Spirtes and Zhang 2006). Split faithfulness into *adjacency faithfulness* (adjacent $\Rightarrow$ no separator) and *orientation faithfulness*: if $X \to Z \leftarrow Y$ is an unshielded collider, X, Y are dependent given every subset of $\mathbf{V} \setminus \{X, Y\}$ containing Z; otherwise every separator contains Z. Algorithmically, check all subsets of X's and of Y's potential parents: if Z is in no separating set, orient a v-structure; if in all, a non-collider; otherwise mark the triple "unfaithful" (shown underlined in the output). This makes the algorithm robust to violations of orientation faithfulness at modest extra cost.

Order-independent PC (Colombo and Maathuis 2014). PC removes an edge as soon as a separator is found; a false-negative CI test early on changes later adjacency sets, so the output depends on the variable ordering (over 25 random orderings the learned skeletons vary). Remedies: average over random orderings, or remove edges only after finishing all tests with the same conditioning-set size; conservative orientation is too restrictive, so a majority rule over separating sets is introduced.

Fast Causal Inference (FCI) relaxes causal sufficiency (allows latent confounders). Example: true graph $X \to Y$, $W \to Z$, and a latent U with $U \to Y$, $U \to Z$. After CI-based edge removal the skeleton is $X - Y - Z - W$; FCI outputs $X \circ\!\!\to Y \leftrightarrow Z \leftarrow\!\!\circ W$, indicating an unmeasured confounder of Y and Z. PAG edge marks:

$X_1 \quad X_2$ (no edge)	X_1 and X_2 are not adjacent
$X_1 \circ\!\!\to X_2$	X_2 is not an ancestor of X_1
$X_1 \circ\!\!-\!\!\circ X_2$	no set d-separates X_1 and X_2
$X_1 \to X_2$	X_1 is a cause of X_2
$X_1 \leftrightarrow X_2$	there is a latent common cause of X_1 and X_2

10.1.2.5 Summary

CI constraints yield partial information about the structure. With a CI oracle (all tests correct), PC is complete (it recovers the full CPDAG). CI testing in practice uses partial correlation (linear Gaussian), Fisher's exact or χ^2 tests (categorical), kernel-based or model-based tests (Appendix 10.1.8). Relaxing sufficiency and/or faithfulness is active research.

10.1.3 Score-based approaches

Motivation. Search for the graph that best fits the data with a complexity penalty.

10.1.3.1 Why scores?

Constraint-based methods rely on CI tests, which can be unreliable in finite samples; a single wrong test can propagate errors through skeleton and orientation. Score-based methods define a function

measuring how well a graph fits the data (and how simple it is) and search over graphs. The paradigms are complementary; hybrid methods (e.g. MMHC) combine them.

Why it is possible. Relative to the true structure (e.g. $E \to A \leftarrow B$, $A \to S$), *adding* an arc increases the number of parameters and encodes wrong causal assumptions; *missing* an arc cannot be compensated by fitting parameters accurately and also misses causal structure.

Intuition. True graph $A \to B \leftarrow C$, proposed $A \to B \to C$. In truth $A \perp\!\!\!\perp C$ but $A \not\perp\!\!\!\perp C \mid B$; the proposed graph says $A \not\perp\!\!\!\perp C$ and $A \perp\!\!\!\perp C \mid B$. Data will likely show $A \perp\!\!\!\perp C$ and $A \not\perp\!\!\!\perp C \mid B$, so the proposal fits poorly. Fit is assessed by the (log-)likelihood.

10.1.3.2 Log-likelihood for DAG models

With $P(\mathbf{V}) = \prod_i P(V_i \mid Pa_{\mathcal{G}}(V_i))$ and models $P_{\theta_i}(V_i \mid Pa_{\mathcal{G}}(V_i))$, the log-likelihood of data $D = \{\mathbf{v}^{(j)}\}_{j=1}^n$ is

$$\ell(\theta; D) = \sum_{j=1}^{n} \sum_{i=1}^{k} \log P_{\theta_i}(v_i^{(j)} \mid \mathbf{pa}_{\mathcal{G}}^{(j)}(V_i)),$$

a *decomposable* likelihood: each θ_i is fitted separately on the part of the data involving $P(V_i \mid Pa_{\mathcal{G}}(V_i))$.

For the true $A \to B \leftarrow C$ and proposed $B \to A$, $C \to B$:

$$\ell_{\text{left}} = \textstyle\sum_j \left[\log P_A(a_j) + \log P_B(b_j \mid a_j, c_j) + \log P_C(c_j)\right],$$
$$\ell_{\text{right}} = \textstyle\sum_j \left[\log P_A(a_j \mid b_j) + \log P_B(b_j \mid c_j) + \log P_C(c_j)\right].$$

Maximized, the true graph has the higher likelihood: with saturated local models, $\ell_{\text{left}}/n - \ell_{\text{right}}/n = I(A; C \mid B) > 0$ (Appendix 10.1.7).

The problem. Compare the true graph with $B \to A$, $C \to A$, $C \to B$ (complete): $\ell_{\text{right}} = \sum_j [\log P_A(a_j \mid b_j, c_j) + \log P_B(b_j \mid c_j) + \log P_C(c_j)]$ fits *better*, because the left model $\{P : A \perp\!\!\!\perp C\}$ is a submodel of the right (all of $P(A, B, C)$). With likelihood alone the best model is always a complete graph: overfitting. Too many parameters fit well but miss the sparse structure; reward fit, but punish parameters.

10.1.3.3 BIC

$$\text{BIC} = -2\,\ell(\hat{\theta}; D) + m \log n,$$

with m the number of parameters; lower is better. For a linear Gaussian network $X_i = \beta_{0i} + \sum_{X_k \in Pa(X_i)} \beta_{ki} X_k + \epsilon_i$, $\epsilon_i \sim \mathcal{N}(0, \sigma_i^2)$, node i contributes $|Pa(X_i)| + 2$ parameters (slopes, intercept, variance), so $m = \sum_i (|Pa(X_i)| + 2) = |E| + 2|V|$. BIC is a regularization over graph structures.

10.1.3.4 Search and GES

Naively listing all DAGs and scoring each is impossible (super-exponentially many DAGs, possibly intractable likelihoods). Local search moves between states (DAGs) by local moves able to reach any DAG, ideally finding the best state in a polynomial sequence of moves. **Greedy Equivalence Search** (GES; Chickering 2002) defines states and moves on equivalence classes:

1. Start with the empty graph.
2. *Addition phase:* repeatedly move to the best class obtained by adding a single edge, until no improvement. (Adding an edge to a class can lead to different classes depending on the representative: adding $B - C$ to $\{A \to B \;\; C,\; A \leftarrow B \;\; C\}$ leads either to $\{A \to B \leftarrow C\}$ or to the class of chains/forks on A, B, C.)

3. *Removal phase:* repeatedly move to the best class obtained by removing a single edge, until no improvement.
4. Return the current class.

Theorem 10.2 (Chickering 2002)

Under faithfulness and as $n \to \infty$, GES identifies the true equivalence class.

Proof [Supplement]. Under faithfulness and with a consistent score such as BIC, two facts hold as $n \to \infty$:

(a) adding an edge that creates a dependence present in P increases the score, and removing an edge whose dependence is absent from P increases the score (local consistency);

(b) the score is equivalent (equal for Markov-equivalent DAGs). The forward phase of GES adds edges while the score improves, and by (a) and Meek's conjecture (proved by Chickering) it stops at an equivalence class whose DAGs contain the true independence model as an I-map. The backward phase then deletes edges while the score improves, and by (a) it can stop only at a minimal I-map, which under faithfulness is the true equivalence class. The detailed argument is Chickering (2002), Theorem 4. ■

Operators (on CPDAGs). Insert$(u, v, \mathbf{T})$: add $u \to v$ for non-adjacent u, v, with $\mathbf{T} \subseteq \mathrm{Ne}(v) \backslash \mathrm{Adj}(u)$ (undirected neighbors of v not adjacent to u); orient every $t \in \mathbf{T}$ as $t \to v$ (possibly creating unshielded colliders); convert to a CPDAG. Delete$(u, v, \mathbf{H})$: remove the edge between u and v, with $\mathbf{H} \subseteq \mathrm{Ne}(v) \cap \mathrm{Adj}(u)$; orient $h \to v$ and $h \to u$ for each $h \in \mathbf{H}$; convert to a CPDAG. Each operator yields a valid CPDAG, so the search stays in class space; at each step GES picks the operator (with its $u, v, \mathbf{T}$ or $\mathbf{H}$) with the best score improvement.

Example 10.3

True DAG $A \to B \leftarrow C$ (a v-structure, a unique class). Forward: start empty; Insert$(A, B, \emptyset)$ gives $A - B$ (undirected, as $A \to B$ and $A \leftarrow B$ are equivalent); Insert$(C, B, \{A\})$ (valid since $A \in \mathrm{Ne}(B) \setminus \mathrm{Adj}(C)$) gives $A \to B \leftarrow C$; no further insertion helps. Backward: no deletion helps. Output $A \to B \leftarrow C$.

10.1.3.5 Continuous optimization: NOTEARS and DAGMA

NOTEARS (Zheng et al. 2018) learns a linear SEM (noise not necessarily Gaussian) by black-box optimization. Example: $X = \epsilon_X$, $Z = aX + \epsilon_Z$, $Y = bX + cZ + \epsilon_Y$, i.e.

$$\begin{pmatrix} X \\ Z \\ Y \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & 0 \end{pmatrix} \begin{pmatrix} X \\ Z \\ Y \end{pmatrix} + \begin{pmatrix} \epsilon_X \\ \epsilon_Z \\ \epsilon_Y \end{pmatrix} \iff \begin{pmatrix} X \\ Z \\ Y \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b + ac & c & 1 \end{pmatrix} \begin{pmatrix} \epsilon_X \\ \epsilon_Z \\ \epsilon_Y \end{pmatrix}.$$

With data $\mathbf{X} \in \mathbb{R}^{n \times d}$ and coefficients $W \in \mathbb{R}^{d \times d}$,

$$\min_{W \in \mathbb{R}^{d \times d}} \frac{1}{2n} \|\mathbf{X} - \mathbf{X}W\|_F^2 + \lambda \|W\|_1 \quad \text{subject to} \quad h(W) = \mathrm{tr}\left(e^{W \circ W}\right) - d = 0.$$

Why this characterizes acyclicity: for a binary adjacency matrix B, B^k counts walks of length k ($B^0 = I$), $e^B = \sum_k B^k / k!$, and $\mathrm{tr}(e^B)$ counts closed walks weighted by $1/k!$, including the d trivial ones; hence $\mathrm{tr}(e^B) = d$ iff B is a DAG. $\mathrm{tr}(e^{W \circ W})$ counts weighted cycles (including 0-length ones) and is smooth, so the problem is solved by an augmented Lagrangian (NOTEARS: Non-combinatorial Optimization via Trace Exponential and Augmented lagRangian for Structure learning).

NOTEARS-MLP (Zheng et al. 2020) learns general mechanisms f: $\min_f L(f)$ subject to $\mathcal{G}(f) \in$ DAGs, with $L(f) = \frac{1}{n}\sum_{j=1}^{d} \ell(x_j, f_j(\mathbf{X}))$, $\mathbb{E}[X_j \mid X_{\mathrm{pa}(j)}] = g_j(f_j(X))$, $\mathbb{E}f_j(X) = 0$, where $k \to j$ is an edge iff f_j depends on X_k. A surrogate for W: f_j is independent of X_k iff $\|\partial_k f_j\|_{L^2} = 0$, so set $[W(f)]_{kj} = \|\partial_k f_j\|_{L^2}$ and solve $\min_{f_j \in H^1(\mathbb{R}^d)} L(f)$ s.t. $h(W(f)) = 0$ (H^1: Sobolev space of square-integrable functions with square-integrable derivatives). With MLPs $\mathrm{MLP}(u; A^{(1)}, \ldots, A^{(h)}) = \sigma(A^{(h)} \cdots \sigma(A^{(2)}\sigma(A^{(1)}u)))$, the output is independent of u_k if the k-th column of $A^{(1)}$ is zero, so the problem becomes

$$\min_{\theta} \ \frac{1}{n}\sum_{j=1}^{d} \ell(x_j, \mathrm{MLP}(\mathbf{X}; \theta_j)) + \lambda \|A_j^{(1)}\|_{1,1} \quad \text{s.t.} \quad h(W(\theta)) = 0,$$

with

$$[W(\theta)]_{kj} = \|k\text{-th column of } A_j^{(1)}\|_2.$$

DAGMA (Bello, Aragam and Ravikumar 2022). The trace-exponential penalty is smooth but numerically demanding (matrix exponentials) and its optimization can be difficult for larger graphs. DAGMA uses a log-determinant penalty

$$h_{\mathrm{DAGMA}}(W) = -\log\det(sI - W \circ W) + d\log s, \qquad s > 0 \ (\text{e.g. } s = 1).$$

Acyclic graphs get zero penalty, graphs with cycles positive penalty, which grows quickly near highly cyclic regions and gives a useful signal before the graph is acyclic. It keeps the NOTEARS setup, optimizing fit, sparsity and acyclicity together, $\min_\theta Q(f_\theta; \mathbf{X}) + \beta_1\|\theta\|_1 + h_{\mathrm{DAGMA}}(W(\theta))$, for linear or nonlinear SEMs, and reports faster optimization with competitive accuracy. It is not a new causal assumption, only another optimization device for enforcing DAGness.

10.1.3.6 Summaries

Score-based. BIC and GES guarantees are large-sample; with finite data correctness is not guaranteed. Local search is intractable if the score cannot be evaluated in polynomial time. If the local models $P(V_i \mid Pa_{\mathcal{G}}(V_i))$ are misspecified, the graph is likely wrong. No hidden variables are assumed.

Constraint- vs. score-based. Constraint-based (PC, FCI): CI tests prune edges, orientation via sepsets and Meek's rules; output a CPDAG (PAG for FCI); sensitive to CI errors. Score-based (GES, NOTEARS, DAGMA): a score is optimized over graph space; tends to do better with small samples, requires correct model specification. Both assume faithfulness and (except FCI) causal sufficiency; hybrids (MMHC) combine both.

10.1.4 Parametric approaches

Motivation. Use functional-form assumptions to orient edges that are indistinguishable from CIs alone.

Under what conditions can we orient $X \to Y$ versus $Y \to X$? With two variables there are no CIs to exploit; asymmetries in the joint distribution must be used.

Additive noise model (ANM). $y = f_y(x) + u_y$ with $x \perp\!\!\!\perp u_y$. If the distribution is known to come from an ANM, compare

$$\text{H1: } y = f_y(x) + u_y,\ x \perp\!\!\!\perp u_y \ (x \to y) \qquad \text{vs.} \qquad \text{H2: } x = f_x(y) + u_x,\ y \perp\!\!\!\perp u_x \ (y \to x).$$

Two main cases:

(1) f non-linear;

(2) f linear but u_y non-Gaussian.

10.1.4.1 Non-linear functions

For non-linear f (generically), an ANM holds in only one direction: in the wrong direction the residual's spread depends on the regressor. Procedure, given (x_i, y_i):

1. Fit both directions with any ML method: $\hat{y} = \hat{f}_y(x)$ and $\hat{x} = \hat{f}_x(y)$.
2. Compute residuals (guesses of u): $\hat{u}_x = x - \hat{f}_x(y)$, $\hat{u}_y = y - \hat{f}_y(x)$.
3. Test residual independence (e.g. HSIC): $\hat{u}_y \perp\!\!\!\perp x$ suggests $X \to Y$; $\hat{u}_x \perp\!\!\!\perp y$ suggests $Y \to X$. If neither holds, the ANM assumption is suspect.

10.1.4.2 Linear functions with non-Gaussian noise: LiNGAM

Cramér's decomposition theorem: if the sum of two independent real random variables is Gaussian, both summands are Gaussian; by induction the same holds for any finite sum. This complements the CLT: sums approach Gaussianity but cannot be exactly Gaussian unless every summand is.

LiNGAM (linear non-Gaussian acyclic model). For $y = ax + u_y$ with non-Gaussian errors,

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ a & 1 \end{pmatrix} \begin{pmatrix} u_x \\ u_y \end{pmatrix}, \qquad x = u_x, \quad y = au_x + u_y.$$

In the backward regression of x on y, the residual is a mixture of u_x and u_y that is uncorrelated with, but (by Darmois–Skitovich/Cramér-type results) not independent of, y; in the forward direction the residual u_y is independent of x.

The trick: ICA. Suppose independent non-Gaussian sources $s_1, \ldots, s_n$ are observed only through a mixture $\mathbf{x} = A\mathbf{s}$. How can A and $\mathbf{s}$ be recovered? By the CLT intuition, $x = as_1 + bs_2$ looks more Gaussian than s_1 or s_2; ICA finds directions maximizing non-Gaussianity. Measures:

- *Kurtosis* (fourth cumulant) $\kappa(y) = \frac{\mathbb{E}[(y-\mu)^4]}{\mathrm{Var}(y)^2} - 3$: zero for Gaussians, positive for super-Gaussian (heavy-tailed, e.g. Laplace), negative for sub-Gaussian (light-tailed, e.g. uniform); simple but outlier-sensitive.
- *Negentropy* $J(y) = H(y_{\text{Gauss}}) - H(y) \geq 0$, with y_{Gauss} Gaussian with the same mean and variance; $J(y) = 0$ iff y is Gaussian (the Gaussian maximizes entropy for a given variance).

ICA takes $\mathbf{x} = A\mathbf{s}$ and returns an unmixing matrix W_{ICA} with $\hat{\mathbf{s}} = W_{\text{ICA}}\mathbf{x}$, up to permutation and scaling (including sign): among all $\mathbf{w}^\top\mathbf{x}$, the maximally non-Gaussian ones are the sources (Hyvärinen and Oja 2000). In the two-variable case, run ICA on (x, y) and look for the zero in the mixing matrix.

LiNGAM (Shimizu et al. 2006). For ordered variables $x_i = \sum_{j<i} b_{ij}x_j + e_i$, i.e. $\mathbf{x} = B\mathbf{x} + \mathbf{e}$, $(I - B)\mathbf{x} = \mathbf{e}$, $\mathbf{x} = (I - B)^{-1}\mathbf{e}$. Let $W = I - B$. ICA gives $W_{\text{ICA}} = PDW$ (P a permutation, D diagonal: the permutation, scaling and sign indeterminacies), but the LiNGAM structure resolves them:

1. $\mathbf{e} = (I - B)\mathbf{x} = W\mathbf{x}$, with W a row-permuted, rescaled version of $I - B$.
2. Run ICA: $W_{\text{ICA}} = \text{ICA}(\mathbf{X})$.
3. Permute the rows of W_{ICA} so that the diagonal has no zeros (the unique such permutation, since $I - B$ has a non-zero diagonal and is permutable to lower-triangular), then rescale rows so the diagonal is one: $W = \text{normalize}(\text{rowpermute}(W_{\text{ICA}}))$.
4. $\hat{B} = I - W$; a causal order is the one making $\hat{B}$ lower-triangular.

Summary. Given an ANM $y = f(x) + u_Y$ with f non-linear or U_Y non-Gaussian, the direction of the arrow is identifiable, assuming no latent confounding ($X \perp\!\!\!\perp U_Y$).

10.1.4.3 Score matching (NoGAM)

(Montagna et al. 2023.) Consider an additive non-linear model $X_i = f_i(\mathrm{PA}_i) + N_i$ with arbitrary noise. The *score function* is the gradient of the log-density, $s(X) = \nabla_X \log p(X)$, $s_i(X) = \partial_{X_i} \log p(X)$ (unrelated to BIC/GES scores): it describes how the log-density changes when one variable is moved. Score matching estimates $s(X)$ from samples without estimating $p(X)$, and the score identifies *leaf* nodes; recursively removing leaves gives a topological order, after which extra edges are pruned.

Why leaves are special. Writing $\log p(X) = \sum_k \log p_k(N_k)$ with $N_k = X_k - f_k(\mathrm{PA}_k)$,

$$s_l(X) = \partial_{N_l} \log p_l(N_l) \quad \text{for a leaf } X_l,$$

$$s_i(X) = \partial_{N_i} \log p_i(N_i) - \sum_{k \in \mathrm{CH}_i} \partial_{N_k} \log p_k(N_k)\, \partial_{X_i} f_k \quad \text{otherwise.}$$

For a leaf, regressing X_l on the rest gives residual $R_l = X_l - q(X_{-l}) = N_l$ (in population), so $s_l(X)$ is a function of R_l alone; for non-leaves the children terms remain and R_i alone generally cannot predict $s_i(X)$. Population statement (under ANM identifiability conditions):

$$X_i \text{ is a leaf} \iff \min_h \mathbb{E}\big[\{h(R_i) - s_i(X)\}^2\big] = 0.$$

Algorithm (NoGAM). (1) Estimate $\hat{s}(X) \approx \nabla_X \log p(X)$.

(2) For each remaining node, regress X_i on X_{-i} and compute residuals $\hat{R}_i$.

(3) Predict $\hat{s}_i(X)$ from $\hat{R}_i$ and record the MSE.

(4) Declare the node with smallest MSE a leaf, remove it, and repeat.

(5) Reverse the removal order to get a topological order, then prune extra edges. This recovers an order without assuming Gaussian noise.

10.1.5 Appendix: GES operators in detail

Insert$(u, v, \mathbf{T})$ adds $u \to v$ (with $u \notin \mathrm{Adj}(v)$), orients $t - v$ as $t \to v$ for every $t \in \mathbf{T}$ ($t \notin \mathrm{Adj}(u)$), and completes the PDAG to a CPDAG. Key set: $\mathbf{T} \subseteq \mathrm{Ne}(v) \setminus \mathrm{Adj}(u)$, and $(\mathrm{Ne}(v) \cap \mathrm{Adj}(u)) \cup \mathbf{T}$ must be a clique; full validity also requires every semi-directed path from v to u to pass through this blocking set. In the v-structure example, after $A - B$ the correct move is $\mathrm{Insert}(C, B, \{A\})$.

Delete$(u, v, \mathbf{H})$ removes the edge between u and v ($u - v$ or $u \to v$), orients $h - v$ as $h \to v$ and $h - u$ as $h \to u$ for every $h \in \mathbf{H}$ (h points into both endpoints), and completes to a CPDAG. Key set: $\mathbf{H} \subseteq \mathrm{Ne}(v) \cap \mathrm{Adj}(u)$, and $(\mathrm{Ne}(v) \cap \mathrm{Adj}(u)) \setminus \mathbf{H}$ must be a clique. These are Chickering's (2002) operators; XGES reparametrises the same score-relevant sets (Nazaret et al. 2025, "Extremely greedy equivalence search").

10.1.6 Appendix: Post-nonlinear causal model (optional)

PNL extends the ANM with an invertible transformation of the effect: for $X_1 \to X_2$, $X_2 = g_2\{f_1(X_1) + E_2\}$, $E_2 \perp\!\!\!\perp X_1$, g_2 invertible. Then $g_2^{-1}(X_2) = f_1(X_1) + E_2$, i.e. $E_2 = g_2^{-1}(X_2) - f_1(X_1)$. Find ψ_2, ϕ_1 such that $E_2 = \psi_2(X_2) - \phi_1(X_1)$ is independent of X_1; if this works in one direction but not the other, orient the edge. Identifiability holds generically, with special functional/noise cases still ambiguous (Zhang and Hyvärinen 2009).

10.1.7 Appendix: Scores in detail

Why the left likelihood is larger. For left $A \to B \leftarrow C$ and right $B \to A$, $C \to B$, with saturated local models the maximized population log-likelihoods are negative conditional entropies:

$$\ell_{\text{left}}/n = -H(A) - H(C) - H(B \mid A, C), \qquad \ell_{\text{right}}/n = -H(C) - H(B \mid C) - H(A \mid B).$$

If the truth is the collider, $A \perp\!\!\!\perp C$ and $H(A) + H(C) + H(B \mid A, C) = H(A, B, C)$, so

$$\ell_{\text{left}}/n - \ell_{\text{right}}/n = H(A \mid B) + H(B \mid C) - H(A) - H(B \mid A, C) = I(A; C \mid B).$$

Conditioning A on B helps the right graph, but dropping A from B's parents hurts more, exactly when $A \not\perp\!\!\!\perp C \mid B$.

Bayesian Dirichlet scores (categorical data). The BD score (Cooper and Herskovits) computes the probability of a network given data, under assumptions. Drawbacks: the user must specify α_{ijk} for every variable–parent configuration, and equivalent structures need not get equal scores. BDe is the score-equivalent variant obtained via an equivalent sample size α; BDeu sets $\alpha_{ijk} = \alpha/(r_i q_i)$, spreading α uniformly over local CPT cells (not a uniform prior over graphs). With i indexing variables, j parent configurations (q_i of them) and k values (r_i of them), and the Dirichlet as conjugate prior of the multinomial,

$$\text{BD}(\mathcal{G}, D; \alpha) = \prod_{i=1}^{N} \text{BD}(X_i \mid Pa_{X_i}; \alpha_i) = \prod_{i=1}^{N} \prod_{j=1}^{q_i} \frac{\Gamma(\alpha_{ij})}{\Gamma(\alpha_{ij} + n_{ij})} \prod_{k=1}^{r_i} \frac{\Gamma(\alpha_{ijk} + n_{ijk})}{\Gamma(\alpha_{ijk})},$$

α_i being the imaginary (equivalent) sample size. $\alpha_{ijk} = 1$ gives the K2 score; $\alpha_{ijk} = 1/2$ the BD score with Jeffreys' prior; $\alpha_{ijk} = \alpha/(r_i q_i)$ the BDeu score (with $\alpha_i = \alpha$ for all X_i).

BIC (maximization form). Dividing by -2,

$$\text{BIC}(\mathcal{G}, D) = \sum_{i=1}^{N} \Big[\ell_i(\hat{\theta}_i) - \frac{\log n}{2} q_i (r_i - 1) \Big], \qquad \ell_i(\hat{\theta}_i) = \sum_{m=1}^{n} \log \hat{P}_i(x_i^{(m)} \mid \pi_i^{(m)}),$$

which is score-equivalent. BIC is the large-sample Laplace approximation of log BDeu; they agree up to $O(1)$ as $n \to \infty$.

10.1.8 Appendix: CI tests for causal discovery

χ^2 test (categorical). Tests $P(X, Y \mid Z) = P(X \mid Z)P(Y \mid Z)$ for all z. Example (2008 General Social Survey, "Where do you get most of your information about current news events?"; credit Yibi Huang):

Age	TV	Internet	Newspapers	Other	Total
18–29	109	92	25	36	262
30–49	272	157	88	63	580
50+	345	59	165	63	632
Total	726	308	278	162	1474

(Did the way people get news change with age?) For each level of Z, expected counts under CI are $E(x, y \mid z) = N(x, z)N(y, z)/N(z)$, and

$$\chi^2(Z = z) = \sum_{x,y} \frac{(O(x, y \mid z) - E(x, y \mid z))^2}{E(x, y \mid z)}, \qquad \chi^2_{\text{total}} = \sum_z \chi^2(Z = z),$$

with $df = (|X| - 1)(|Y| - 1)|Z|$; the p-value compares χ^2_{total} with the χ^2_{df} distribution.

Partial correlation (linear).

$$r_{XY\cdot Z} = \frac{r_{XY} - r_{XZ}r_{YZ}}{\sqrt{1-r_{XZ}^2}\sqrt{1-r_{YZ}^2}}$$

for a single conditioning variable (for multivariate Z use recursion or regression residuals). If it remains high, X and Y are correlated beyond Z; if it drops, Z explains most of their correlation.

HSIC (non-linear marginal independence). Tests $H_0 : X \perp\!\!\!\perp Y$ without assuming linearity. With kernels k, l, Gram matrices K, L and $H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$, $\widehat{\mathrm{HSIC}}(X,Y) = \frac{1}{n^2}\operatorname{tr}(KHLH)$. With characteristic kernels, population HSIC is zero iff $X \perp\!\!\!\perp Y$. Null calibration by permutation, gamma approximation or asymptotics (Gretton et al. 2005). Useful when marginal dependence may be non-linear and non-Gaussian; for conditional independence use KCI.

Kernel conditional independence test (KCI). Targets $H_0 : X \perp\!\!\!\perp Y \mid Z$ via RKHS conditional covariances: represent X, Y, Z by kernels K_X, K_Y, K_Z; regress out Z in the RKHS, e.g. $R_Z = \epsilon(K_Z + \epsilon I)^{-1}$, $\tilde{K}_X = R_Z K_X R_Z$, $\tilde{K}_Y = R_Z K_Y R_Z$; test whether the residualised dependence $T_{\mathrm{KCI}} = \frac{1}{n}\operatorname{tr}(\tilde{K}_X\tilde{K}_Y)$ is zero. It replaces Fisher-z or χ^2 tests in PC-style discovery when non-linear CI matters, at higher computational cost and with sensitivity to kernel bandwidth and regularization (Zhang, Peters, Janzing and Schölkopf 2011).

10.2 Local causal discovery

Motivation. The central question is what happens to structure learning when the reasoning is restricted to local objects (Markov blankets, a target's neighborhood, order prefixes) instead of the whole graph.

10.2.1 The map

Motivation. Organize discovery algorithms by signal and reach.

The previous lecture gave the global workhorses: constraint-based (PC, FCI), using CI tests over the whole graph, and score-based (GES, NOTEARS, DAGMA), optimizing one global score. This lecture is not a new family but a direction: *what happens when we go local?* Two knobs organize everything:

- **Signal** (vertical): test independences or optimize a score?
- **Reach** (horizontal): reason over the whole graph or a local object?

	global	local
CI tests	PC, FCI	GS, HITON
CI → score		MMHC
score	GES, DAGMA	BOSS

The previous lecture filled the global column; this one fills the local column.

10.2.1.1 One primitive: grow–shrink

A recurring two-step move, reused on different local objects:

- **Grow:** add while the candidate still helps. CI version: add Y while $X \not\perp\!\!\!\perp Y \mid \mathbf{S}$; score version: add Y while the score improves.

- **Shrink:** remove what became redundant. CI version: drop Y if $X \perp\!\!\!\perp Y \mid \mathbf{S} \setminus \{Y\}$; score version: drop Y if the score improves.

Algorithm	Local object	Signal
GS	Markov blanket	CI
HITON	one target's parents and children	CI
MMHC	skeleton, then DAG	CI + score
BOSS	order prefix	score

10.2.2 Going local on CI

Motivation. Learn Markov blankets and target neighborhoods by CI tests.

10.2.2.1 The Markov blanket

For a node X,

$$\mathrm{MB}(X) = \big(\mathrm{pa}(X) \cup \mathrm{ch}(X) \cup \mathrm{pa}(\mathrm{ch}(X))\big) \setminus \{X\},$$

its parents, children and spouses (co-parents of children). Conditioning on $\mathrm{MB}(X)$ makes every other variable irrelevant for X. It is small and local, yet enough to drive skeleton and orientation (Margaritis and Thrun 1999; Margaritis 2003).

10.2.2.2 GS: grow–shrink the blanket

Grow (permissive): add any Y with $X \not\perp\!\!\!\perp Y \mid \mathbf{S}$; some additions are real, others artefacts of an incomplete $\mathbf{S}$. **Shrink** (clean up): drop Y once a better separator appears, $X \perp\!\!\!\perp Y \mid \mathbf{S} \setminus \{Y\}$. Example: in the chain $X \to C \to G$, $X \not\perp\!\!\!\perp G$, so grow may add G; once C is in $\mathbf{S}$, $X \perp\!\!\!\perp G \mid C$ and shrink removes G. Then test which blanket members are true neighbors and orient unshielded colliders as in PC.

Algorithm 2 GS-MB(X)

1: $\mathbf{S} \leftarrow \emptyset$
2: **repeat** find $Y \in \mathbf{V} \setminus (\{X\} \cup \mathbf{S})$ with $X \not\perp\!\!\!\perp Y \mid \mathbf{S}$; if found, $\mathbf{S} \leftarrow \mathbf{S} \cup \{Y\}$
3: **until** no such Y exists
4: **repeat** find $Y \in \mathbf{S}$ with $X \perp\!\!\!\perp Y \mid \mathbf{S} \setminus \{Y\}$; if found, $\mathbf{S} \leftarrow \mathbf{S} \setminus \{Y\}$
5: **until** no such Y exists
6: **return S**

Correctness sketch. Under faithfulness, a parent or child Y of X stays dependent on X given any set not containing X, Y, so grow includes it. If Y is a parent of a child of X, conditioning on that child creates dependence, so grow eventually includes Y. Thus after grow $\mathrm{MB}(X) \subseteq \mathbf{S}$. If $W \notin \mathrm{MB}(X)$ but $W \in \mathbf{S}$, the true blanket (itself a separating set) separates X from W; since $\mathrm{MB}(X) \subseteq \mathbf{S} \setminus \{W\}$, shrink removes W via $X \perp\!\!\!\perp W \mid \mathbf{S} \setminus \{W\}$.

Structure induction. (1) Compute $\mathrm{MB}(X)$ for every X.

(2) For each $Y \in \mathrm{MB}(X)$ decide whether it is a direct neighbor: no edge if $\exists \mathbf{S} \subseteq \mathbf{R}$ with $X \perp\!\!\!\perp Y \mid \mathbf{S}$, where $\mathbf{R}$ is the smaller of $\mathrm{MB}(X) \setminus \{Y\}$ and $\mathrm{MB}(Y) \setminus \{X\}$.

(3) Orient unshielded colliders using local tests.

(4) Repair cycles and propagate forced directions.

Randomized GS. Plain GS becomes expensive with large blankets. Randomized GS samples conditioning sets from the smaller blanket instead of enumerating all subsets and combines multiple tests as evidence for edge existence and direction: an anytime-style algorithm (more tests, more stable decisions) with controlled runtime but weaker exactness guarantees.

10.2.2.3 HITON: target-focused, not whole-graph

GS computes a blanket for every node to assemble the full network. HITON wants only the local neighborhood of one target T (e.g. feature selection for a disease variable). Its workhorse HITON-PC$(\cdot)$ returns a node's *parents and children* ("PC" here is not the PC algorithm) by interleaved grow-while-shrink with small conditioning sets, which is sample-efficient. "Focus" means running this subroutine on T and a bounded local ring around it, not on all n variables: local, not literally single-node. Two pieces: HITON-PC (one node's neighbors) and HITON-MB (assembling T's blanket) (Aliferis et al. 2003; 2010).

HITON-MB: two rings, then prune. Example: $P \to T \to C \leftarrow Z$, $C \to G$.

1. Ring 1: PC$(T) \leftarrow$ HITON-PC(T), T's parents and children (P, C).
2. Ring 2 (PC-of-PC): run HITON-PC(Y) for each $Y \in$ PC(T). A spouse Z is not adjacent to T, so it only surfaces here, as a neighbor of a child.
3. Prune: drop ring-2 candidates still separable from T (e.g. the grandchild G, separated by C). Ring 2 keeps the spouse Z (no set separates it from T once the collider C is conditioned on) and prunes G.

This calls HITON-PC on several nodes, but only on T's local ring. The PC step finds direct

Algorithm 3 HITON-MB(T)

PC$(T) \leftarrow$ HITON-PC(T); $\mathbf{C} \leftarrow$ PC(T)
for each $Y \in$ PC(T) **do** $\mathbf{C} \leftarrow \mathbf{C} \cup$ HITON-PC(Y)
for each candidate $X \in \mathbf{C} \setminus$ PC(T) **do**
    if $\exists \mathbf{S} \subseteq \mathbf{C} \setminus \{X\}$ with $T \perp\!\!\!\perp X \mid \mathbf{S}$ **then** remove X from $\mathbf{C}$
return C

neighbors of the target; the PC-of-PC search brings in co-parents; CI tests remove false positives.

Halfway: the CI column is filled (GS: a blanket per node; HITON: one target). The score-local corner remains: MMHC and BOSS.

10.2.3 Bringing in the score

Motivation. Combine local CI skeletons with scores, and search over orders.

10.2.3.1 MMHC: a hybrid bridge

Start at CI-local, then switch to score (Tsamardinos, Brown and Aliferis 2006). Run local parents-and-children discovery for every node with MMPC, the max-min sibling of HITON-PC (same grow/shrink, different admission rule); the union is a candidate skeleton. A score-based hill-climb then picks a DAG but may add edges only inside that skeleton:

$$\text{local PC per node (MMPC)} \;\to\; \text{union} = \text{skeleton} \;\to\; \text{score hill-climb} \;\to\; \text{one DAG}.$$

It combines a sound skeleton restriction with heuristic score-based orientation: the first time a score drives the graph.

10.2.3.2 BOSS: best order score search

Score-local (Andrews et al. 2023). The search state is a permutation π, not a graph. Each v's candidate parents are its *prefix*, the variables before it:

$$\mathrm{pre}_\pi(v) = \{w : \pi.\mathrm{index}(w) < \pi.\mathrm{index}(v)\};$$

e.g. for $\pi = (A, C, B, D, E)$, $\mathrm{pre}_\pi(B) = \{A, C\}$.

Scoring an order. The projection $\mathcal{G}_\pi$ picks each v's parents from its prefix by greedy grow/shrink, approximating

$$\mathrm{pa}_\pi(v) \approx \underset{\mathbf{W} \subseteq \mathrm{pre}_\pi(v)}{\arg\max}\ \mathrm{score}_v(\mathbf{W}), \qquad \mathrm{score}_v(\mathbf{W}) = \mathrm{BIC}(X_v \mid X_{\mathbf{W}}),$$

a penalized log-likelihood (higher = better fit with fewer parameters). The order score sums the local terms, $\mathrm{score}(\pi) = \sum_v \mathrm{score}_v(\mathrm{pa}_\pi(v))$. Grow–shrink trees (GSTs) cache repeated prefix scores; this is the key to BOSS's scaling. It relies on the local decomposition $\mathrm{BIC}(\mathcal{G}, D) = \sum_{v \in \mathbf{V}} \mathrm{BIC}(X_v \mid X_{\mathrm{pa}_\mathcal{G}(v)})$: changing v's parents affects only one local term, which makes projection and best-move scoring efficient.

Algorithm 4 project$(\mathbf{X}, \pi)$ and best-move$(\mathcal{T}, \pi, v)$

$E \leftarrow \emptyset$
for each v in π **do**
  $\mathbf{Z} \leftarrow \mathrm{pre}_\pi(v)$; $\mathbf{W} \leftarrow \mathrm{grow}(\mathbf{X}, v, \mathbf{Z})$; $\mathbf{W} \leftarrow \mathrm{shrink}(\mathbf{X}, v, \mathbf{W})$
  add $w \to v$ to E for each $w \in \mathbf{W}$
return $\mathcal{G}_\pi = (\mathbf{V}, E)$

best $\leftarrow \mathcal{T}.\mathrm{score}(\pi)$
for $i = 1, \ldots, |\pi|$ **do**
  $\pi' \leftarrow \pi.\mathrm{move}(v, i)$
  if $\mathcal{T}.\mathrm{score}(\pi') >$ *best* **then** *best* $\leftarrow \mathcal{T}.\mathrm{score}(\pi')$; $\pi \leftarrow \pi'$
return π

Example 10.4 (Best-move)

Truth: the collider $A \to C \leftarrow B$. Move one variable to every position, re-project, and keep the move only if $\mathrm{score}(\mathcal{G}_{\pi'}) > \mathrm{score}(\mathcal{G}_\pi)$. With $\pi = (A, C, B)$, C precedes B, so C is a candidate parent of B; conditioning on the collider C couples A and B, so the projection includes an extra (spurious) A–B dependence: 3 edges. Moving C last, $\pi' = (A, B, C)$, gives the true collider with 2 edges. The independences represented are those of the true model with fewer parameters, so $\mathrm{score}(\mathcal{G}_{\pi'}) > \mathrm{score}(\mathcal{G}_\pi)$ and π' is kept.

Benchmarks. BOSS is designed for large, relatively dense models. In reported simulations it matches GRaSP (an earlier order-based search) in accuracy while scaling far better in runtime, compares favorably with fast GES, PC, DAGMA and LiNGAM, and has been demonstrated on resting-state fMRI with hundreds of cortical parcels (dense graphs, hundreds to 1000 variables).

10.2.4 Zoom out

Motivation. Choose among the methods and remember what none of them solve.

The filled map. Going local keeps the signal but shrinks the object; MMHC bridges CI and score; GSTs make the score-local corner scale.

What to use when.

- Sparse graph, reliable CI tests: GS (full graph) or HITON (one target).
- High-dimensional variable/feature selection: HITON's target blanket.
- Large, dense, score-friendly data: BOSS (or MMHC as a simpler hybrid).
- Hidden confounders: none of these; use FCI-type methods.
- Only one effect is needed, not the whole graph: effect-specific local discovery (Entner et al.; LDECC).

What none of these solve / limitations checklist.

- Causal sufficiency: GS, HITON, MMHC and BOSS all assume no hidden confounders; none replaces FCI.
- Faithfulness: separations must mirror independences (or score consistency must hold).
- Finite samples: weak effects and large conditioning sets hurt every method.
- Specification and tuning: the score family (BOSS needs an appropriate local likelihood) and the penalty (e.g. BIC) set the precision–recall balance between sparse and dense recovery.

Locality changes the cost of search, not the assumptions.

Comparison.

	GS	HITON	MMHC	GES	BOSS
Main signal	CI tests	CI tests	CI + score	BIC-like score	BIC-like score
Search object	blankets	target blanket	skeleton + DAG	CPDAGs	orders
Locality	Markov blankets	PC/co-parents	local skeleton	edge operators	prefixes
Output target	MEC graph	variables for T	one DAG	CPDAG	CPDAG
Main risk	CI error	target scope	heuristic orient.	many score calls	model spec.

Summary. One primitive, grow–shrink, on four local objects: GS (CI on Markov blankets), HITON (CI on one target's neighborhood), MMHC (a CI skeleton restricting a score hill-climb), BOSS (score on order prefixes, with GSTs caching repeated work). The same signals as before, made local: cheaper search, same assumptions.